Wage Composition Bias During the Pandemic and Post-Pandemic Inflationary Period in the United States: Evidence from CPS Microdata, 2016-2024

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Abstract

During the COVID-19 pandemic, U.S. employment collapsed while measured wages rose sharply, a puzzling pattern if interpreted as reflecting individual wage gains. Using CPS microdata from 2016–2024, I quantify composition bias through three complementary approaches: demographic adjustment, imputation bounds for non-employed workers, and panel methods with individual fixed effects. Composition bias is substantial and operates in opposing directions depending on the comparison group. Continuously employed men experienced 15.3 percent real wage growth from 2016 to 2022, while cross-sectional measures showed only 5.4 percent, capturing roughly one-third of job-stayer gains. Yet cross-sectional measures also overstate growth relative to the full working-age population; imputation bounds show only 2.6 to 2.8 percent. Observable demographics explain approximately one-third of these composition effects. The negative correlation between cross-sectional wages and employment largely disappears in panel data, confirming that compositional shifts rather than individual wage dynamics account for measured wage volatility. Additionally, lower percentiles of within-cell wage distributions grew faster than higher percentiles, revealing wage compression even among workers with identical observable characteristics.

Section 1: Introduction

The COVID-19 pandemic triggered unprecedented disruption in U.S. labor markets. Employment fell by 22 million workers from February to April 2020, and the employment-to-population ratio for prime-age men declined from 85 percent to 71 percent. Concurrent with this collapse, measured wages rose sharply: in the CPS microdata examined here, cross-sectional median hourly wages increased from $21.70 in early 2020 to $24.70 by mid-year, an increase of approximately 14 percent. This simultaneous employment collapse and wage rise is puzzling, as standard models predict that negative labor demand shocks reduce wages, or at most leave them unchanged under wage rigidity. This paper asks: how much of measured wage volatility during the pandemic reflects actual wage changes for individual workers versus changes in the composition of the employed workforce?

The answer matters for interpreting aggregate wage statistics. If job losses concentrate among lower-wage workers, the median wage among remaining workers rises mechanically, creating an upward composition bias. A rising average wage would then reflect worker exit rather than broadly shared gains, providing a misleading signal about worker welfare. Conversely, as displaced workers return, measured wages may fall despite continued wage increases for those continuously employed. Understanding the magnitude and direction of composition bias is therefore essential for assessing how the pandemic affected workers' economic well-being.

I quantify composition bias using CPS microdata from 2016–2024 through three complementary approaches: demographic adjustment that reweights the employed sample to hold observable characteristics constant, imputation bounds that extend the analysis to include non-employed individuals, and panel methods that track the same workers over time. Three findings emerge. First, composition bias is substantial and operates in opposing directions depending on the comparison group. Continuously employed men experienced 15.3 percent cumulative real wage growth from February 2016 to February 2022, while cross-sectional measures showed only 5.4 percent, capturing roughly one-third of job-stayer gains. Yet cross-sectional measures also overstate wage growth relative to the full working-age population; imputation bounds show only 2.6 to 2.8 percent growth. Second, observable demographics explain only about one-third of the gap between cross-sectional and panel estimates, indicating substantial selection on unobservable characteristics. Third, lower percentiles of within-cell wage distributions grew faster than higher percentiles, revealing wage compression even among workers with identical observable characteristics. This pattern complements Autor, Dube, and McGrew's (2023) finding of aggregate compression during the post-pandemic period.

The literature on composition bias in wage measurement has established that compositional shifts can substantially distort measured wage dynamics. Solon, Barsky, and Parker (1994) show that composition bias in aggregate wage statistics obscures the true cyclicality of wages, finding that individual wage growth is more responsive to economic conditions than cross-sectional measures suggest. Elsby, Shin, and Solon (2016) confirm that composition bias remains substantial during the Great Recession and demonstrate that longitudinal methods are necessary to isolate genuine wage changes from compositional shifts. Most directly relevant to the present study, Cajner et al. (2020) use administrative payroll data to show that low-wage workers disproportionately exited employment during the early pandemic, with more than 35 percent of bottom-quintile workers losing jobs by mid-April 2020. They find that after controlling for worker fixed effects, the sharp rise in measured wages was entirely due to compositional changes, as individual base wages remained flat. This paper extends prior work by systematically characterizing composition bias across multiple comparison groups and time horizons, quantifying the share attributable to observable versus unobservable characteristics, and documenting within-cell wage compression as an additional feature of pandemic-era wage dynamics.

Labor market theories suggest multiple channels through which the pandemic may have generated compositional shifts in employment. In search-and-matching frameworks, negative shocks raise the reservation productivity threshold for continuing matches, causing workers at the lower end of the productivity and wage distribution to exit disproportionately. The pandemic intensified and complicated these forces. Sector-specific demand collapses concentrated job losses in contact-intensive industries that employ disproportionately lower-wage workers (Mongey, Pilossoph, and Weinberg 2021). School and daycare closures forced some workers to withdraw for caregiving responsibilities (Doepke et al. 2022). Enhanced unemployment benefits under the CARES Act may have raised reservation wages, making non-employment temporarily more attractive for workers whose previous earnings fell below the benefit threshold. These multiple selection channels operated simultaneously during the pandemic: involuntary layoffs, caregiving withdrawals, and benefit-induced exits.

However, the CPS data used here cannot distinguish among these mechanisms. I observe employment status and wages but not the reasons for employment transitions or the counterfactual wages workers would have earned had they remained employed. This limitation motivates the paper's descriptive approach: rather than testing which mechanism drove compositional shifts, I quantify the overall magnitude of composition bias and characterize how it varies across comparison groups and methods.

The remainder of this paper proceeds as follows. Section 2 describes the data, sample construction, and methodological approaches. Section 3 presents the main results. Section 4 discusses robustness. Section 5 concludes.

Section 2: Data and Methodology

2.1 Data and Sample Construction

This analysis uses data from the Current Population Survey (CPS) for January 2016 through December 2024. I restrict the sample to men aged 25–54, a group with relatively stable labor force attachment, and exclude individuals in group quarters and active military personnel. For wage analysis, I further exclude observations with missing or allocated wage data. All wage measures are deflated to real January 2016 dollars using the Consumer Price Index for All Urban Consumers (CPI-U).

The CPS employs a rotating panel structure in which households are interviewed for four consecutive months, exit the sample for eight months, and then return for four final months. Outgoing rotation groups (months-in-sample 4 and 8) receive detailed earnings questions, including hourly wages for those paid by the hour and usual weekly earnings for salaried workers. For salaried workers, I construct hourly wages by dividing usual weekly earnings by usual hours worked per week, following Lemieux (2006). All estimates use CPS survey weights to ensure national representativeness.

2.2 Wage Index Construction

I normalize all wage indices to January 2016, when the U.S. economy had largely recovered from the Great Recession and unemployment stood at 4.8 percent. This baseline provides four years of pre-pandemic data to establish wage trends before the 2020 shock. To reduce sampling variability, I apply three-month centered moving averages to all wage series.

2.3 Cross-Sectional Wage Measures

The baseline cross-sectional approach calculates weighted mean and median real hourly wages for employed workers in each month. The median is computed as the weighted 50th percentile to account for the CPS sampling design. These measures are normalized to January 2016 and smoothed as described above. Cross-sectional measures provide a natural baseline but conflate individual wage changes with shifts in workforce composition.

2.4 Demographic Adjustment

To assess how much of the composition effect observable characteristics can explain, I construct composition-adjusted measures that reweight the employed sample each month to match the January 2016 distribution of age and education. I define 30 demographic cells by age (six five-year groups from 25–54) and education (five categories: less than high school, high school diploma, some college or associate's degree, bachelor's degree, and graduate degree). I calculate mean wages within each cell and aggregate using fixed January 2016 population shares. Demographic adjustment isolates wage changes from demographic shifts in the employed workforce. However, it still compares different individuals each month and therefore does not account for selection into employment based on unobservable characteristics.

2.5 Imputation Bounds for Non-Employed Workers

Cross-sectional measures based only on employed workers can misrepresent population-wide wage trends during periods of large employment changes. To address this, I impute wages for non-employed individuals at the 33rd and 50th percentiles of the wage distribution within their demographic cell, following the partial identification logic of Manski (1989, 1994).

The 50th percentile assumption treats non-employed individuals as similar to the median worker in their demographic group, serving as a neutral benchmark. The 33rd percentile assumption places non-employed individuals in the lower portion of their group's wage distribution, reflecting potential negative selection into non-employment. Together, these bounds bracket plausible values for population-wide wage growth.

As with the demographic adjustment method, I fix the population weights for each demographic cell at their January 2016 levels. This ensures that measured wage changes reflect within-cell wage dynamics rather than shifts in the relative size of demographic groups over time. Comparing the two imputation bounds also provides information about how different points of the within-cell wage distribution evolved, which I discuss in Section 3.3.

2.6 Panel Matching with Individual Fixed Effects

The panel approach takes advantage of the CPS longitudinal structure to track individual wage changes, holding workforce composition constant by following the same workers over time. Outgoing rotation groups provide earnings data at months-in-sample 4 and 8, creating 12-month intervals between wage observations. I match individuals across their two outgoing rotation interviews using CPS household and person identifiers. For each successfully matched pair, I calculate the log wage change:

$$\Delta log w_i = log w_{i,t+12} - log w_{i,t}$$

where $w_{i,t}$ is individual i's real hourly wage at the first interview and $w_{i,t+12}$ is their wage twelve months later. This represents the individual i's year-over-year percentage wage growth. For each calendar month t, I compute the weighted mean log wage change across all matched individuals, where $weight_i$ is individual i's CPS survey weight:

$$\Delta log w_t = (\Sigma_i \Delta log w_i \times weight_i) / \Sigma_i weight_i$$

Because CPS earnings data are available only at 12-month intervals, I convert year-over-year changes to monthly equivalents to construct a continuous index comparable to the monthly cross-sectional series. If wages grew by a factor of G over twelve months, the monthly-equivalent growth factor is $\sqrt[12]{G}$:

$$g_t = \sqrt[12]{G_t} = \sqrt[12]{exp(\Delta log\ w_t)}$$

This assumes wage growth occurred smoothly over the twelve-month period. I then construct the cumulative index by chaining these monthly growth factors forward from the January 2016 baseline:

$$Index_t = Index_{t-1} \times g_t$$

with $Index_{jan\ 2016} = 1$. Finally, I apply three-month centered moving averages smoothing:

$$3MA\ Index_t = (Index_{t-1} + Index_t + Index_{t+1})/3$$

The panel method has two limitations. First, it tracks only workers who remain employed in the CPS sample for two consecutive outgoing interviews; those who exit employment or leave the survey are excluded. Second, the panel index extends only through February 2022, the last month for which twelve-month-ahead matches are available in the current data.

A potential concern is that the composition of continuously employed workers may itself shift over time. Table 1 presents summary statistics for the panel sample across years. The demographic composition remains relatively stable: average age varies only between 38.3 and 38.8 years, and while the college share rises from 39.4 to 43.5 percent, this shift is gradual. Because the panel method computes year-over-year wage changes for the same individuals before aggregating, it inherently controls for time-invariant individual characteristics. The observed cumulative growth therefore reflects wage gains for comparable workers rather than shifts in sample composition.

Table 1: Summary Statistics of the Panel Sample (Continuously Employed Workers), 2016-2022

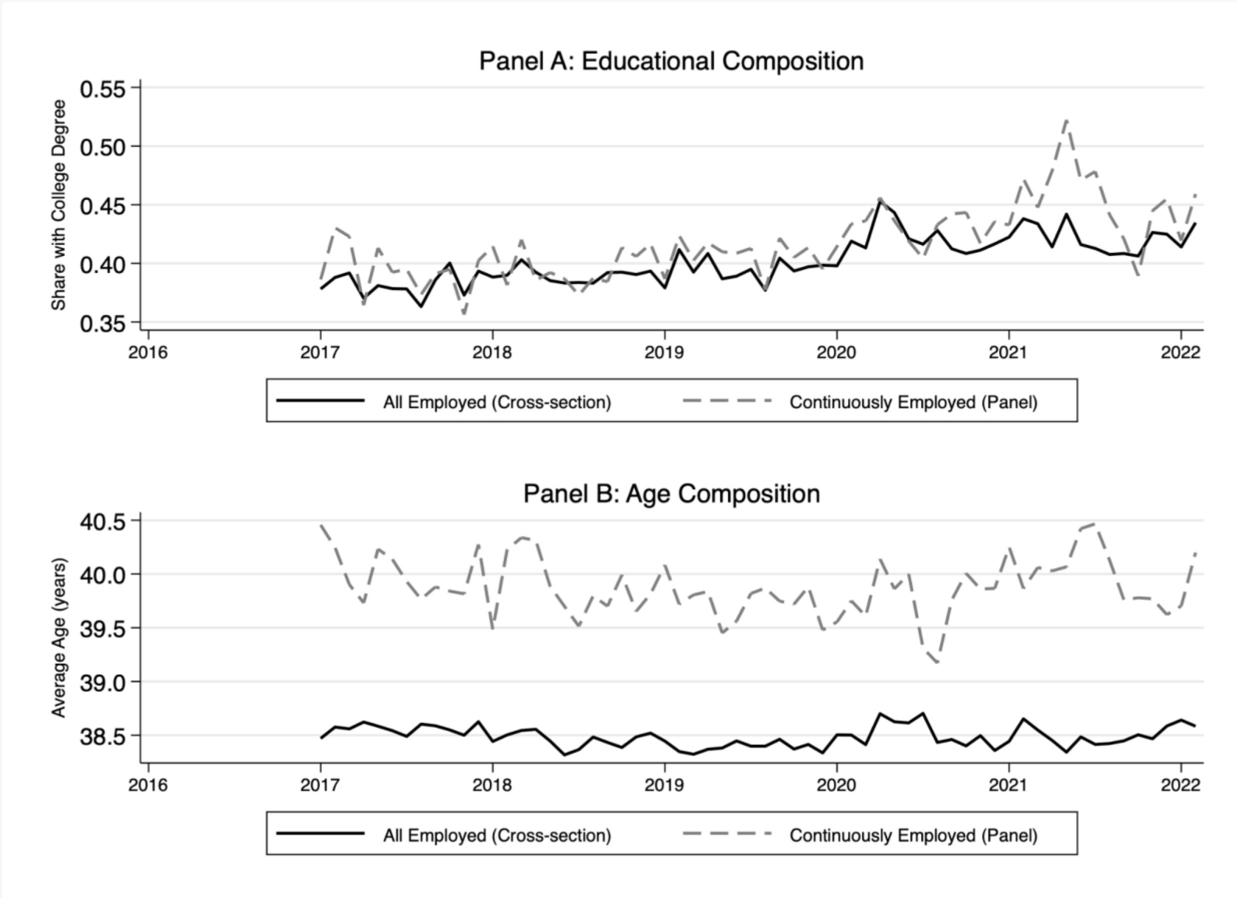
	2016	2017	2018	2019	2020	2021	2022
Demographics:							
Average age (years)	38.8	38.6	38.5	38.3	38.5	38.7	38.6
% with college degree	39.4	38.8	39.0	41.2	41.9	43.8	43.5
Wage distribution (2016$):							
Mean hourly wage	24.85	25.15	24.92	25.32	26.15	27.01	25.67

Standard deviation	20.74	16.35	15.98	16.77	16.02	16.33	14.72
25th percentile	12.99	13.71	13.45	14.06	14.65	14.86	15.03
Median	19.98	20.67	19.60	20.15	21.52	22.52	20.88
75th percentile	31.97	32.73	32.52	33.79	33.89	35.75	32.92
Sample Size	4047	4085	3904	3696	3715	3326	3324

Notes: Sample includes men aged 25–54 matched across consecutive outgoing rotation interviews (months-in-sample 4 and 8), 12 months apart. College degree includes bachelor's degree and above. Wages are deflated to January 2016 dollars using CPI-U. Source: CPS 2016–2022.

Figure 1 compares demographic characteristics of the panel sample to the cross-sectional sample. Continuously employed workers have modestly higher education levels than all employed workers in each month, consistent with positive selection into continuous employment. Following Solon, Barsky, and Parker (1994), such selection likely extends beyond observable characteristics: workers who remain employed through economic disruptions tend to have more stable employment relationships and stronger labor market attachment. This selection does not undermine the comparison; rather, the gap between panel and cross-sectional measures reflects the contribution of workforce churning to measured wage changes.

Figure 1: Panel vs. Cross-Sectional Sample Demographic Characteristics



Notes: Panel sample includes men aged 25–54 matched across 12-month intervals who remained employed at both interviews. Cross-sectional sample includes all employed men aged 25–54 in each month. College share is the proportion with a bachelor's degree or above. Source: CPS 2016–2022.

Section 3: Results

3.1 Main Findings

Figure 2 presents real wage indices from February 2016 through February 2022, comparing the unadjusted cross-sectional mean with four methods addressing composition bias: demographic adjustment, two imputation bounds, and a panel approach with individual fixed effects.

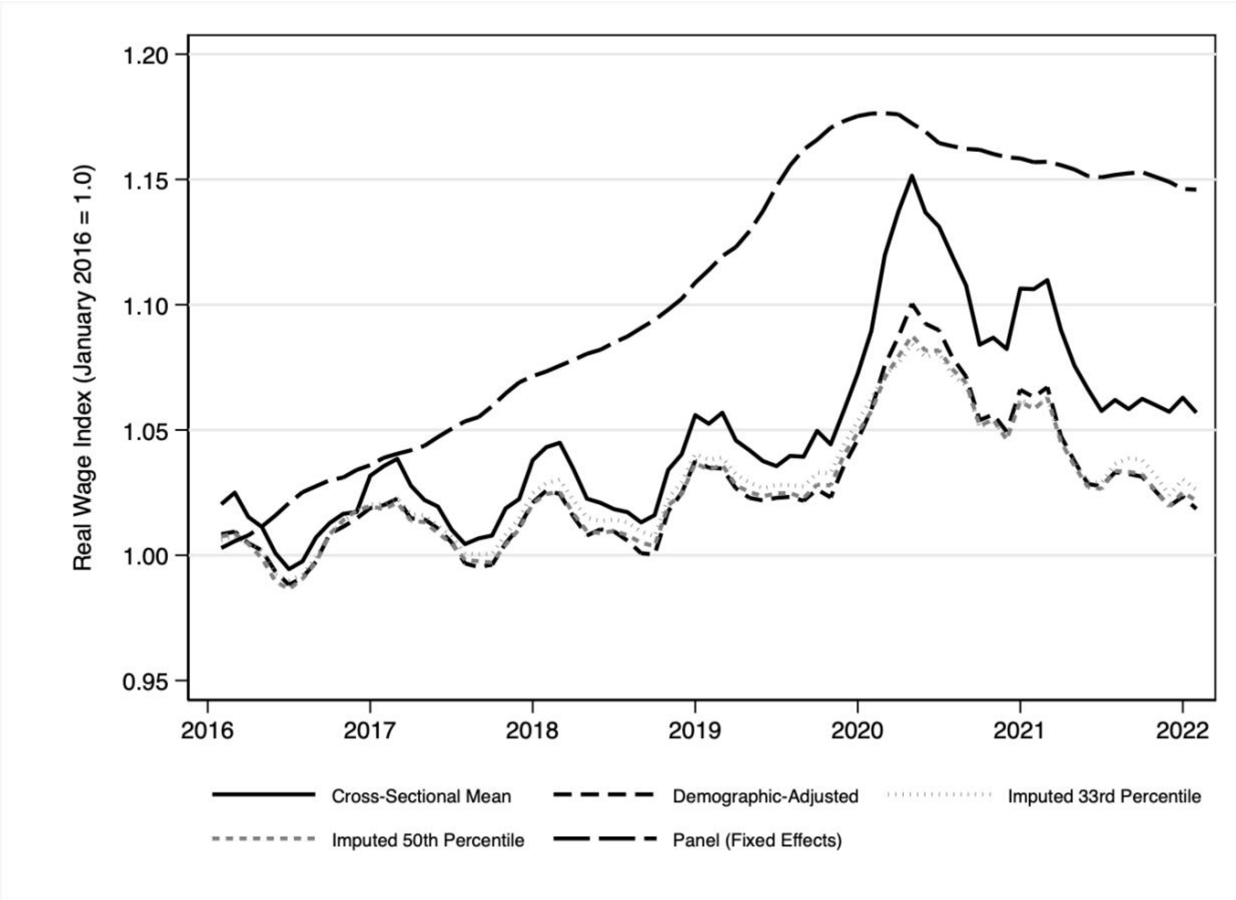
The central finding is the substantial divergence between panel and cross-sectional estimates. Continuously employed workers experienced 15.3 percent cumulative real wage growth over this period, while the cross-sectional mean shows 5.4 percent growth. This 9.9 percentage point gap indicates that cross-sectional measures capture only about one-third of the wage gains experienced by job-stayers.

The demographic adjustment method yields 2.0 percent growth, 3.4 percentage points lower than the unadjusted cross-sectional measure. This indicates that the composition of employed workers shifted toward groups with characteristically higher wages, with the observed demographic shift accounting for approximately 63 percent of unadjusted cross-sectional growth. This pattern is consistent with evidence that employment fell disproportionately among less-educated workers during the pandemic (Cajner et al. 2020; Mongey, Pilossoph, and Weinberg 2021). Figure 3 confirms this pattern in the present data: by April 2020, employment among workers without a high school diploma had fallen approximately 25 percent relative to February, while employment among college-educated workers fell approximately 10 percent.

The 13.3 percentage point gap between demographic-adjusted wages and panel estimates indicates substantial selection on characteristics not captured by age and education. Several factors likely contribute. First, industry and occupation vary in exposure to pandemic disruptions, with workers in contact-intensive service occupations facing elevated layoff risk regardless of demographic characteristics (Mongey, Pilossoph, and Weinberg 2021). Second, firm characteristics matter: workers at larger, more financially stable firms were more likely to retain employment through furlough programs or remote work arrangements. Third, within demographic cells, workers with stronger employment histories, more stable tenure, and higher productivity were likely better positioned to retain jobs. The panel approach implicitly conditions on these unobserved factors by tracking the same individuals but cannot decompose their relative contributions.

Imputation-based estimates that include non-employed individuals show growth of 2.6 to 2.8 percent, indicating that cross-sectional measures overstate wage gains relative to the full working-age population. The comparison between these bounds also reveals information about within-cell distributional changes, which I discuss in Section 3.3.

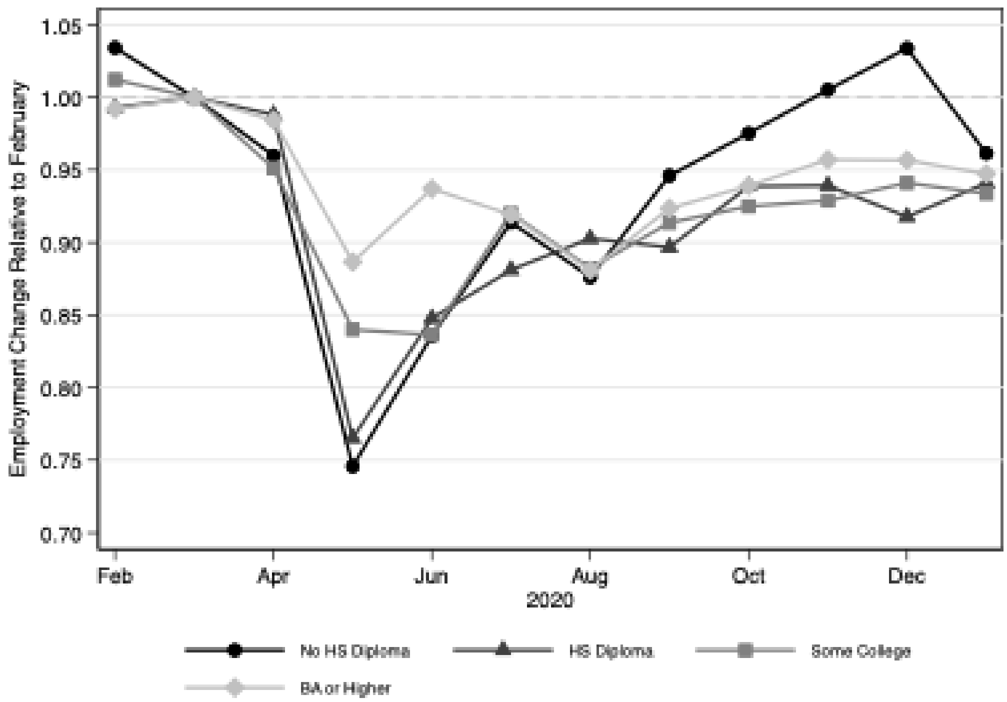
Figure 2. Real Wage Growth of U.S. Men Aged 25-54 During the Pandemic and Post-Pandemic, Controlling for Composition Bias, 2016 - 2022



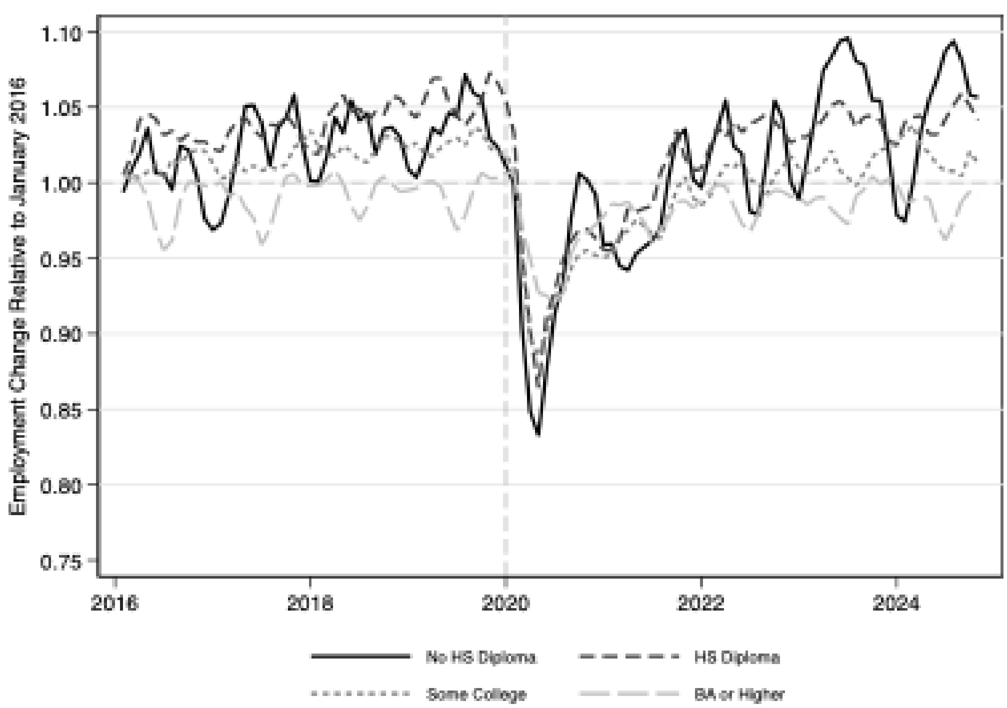
Notes: figure presents real wage indices normalized to January 2016 = 1.0 (values above 1.0 indicate cumulative real wage growth, e.g., 1.15 = 15% growth), smoothed using 3-month centered moving averages. Cross-Sectional Mean includes all employed workers each month. Demographic-Adjusted reweights employed workers to hold age-education distribution constant at 2016 levels. Imputed 33rd/50th Percentile include both employed and non-employed individuals, assigning wages to non-workers at the specified percentile of their demographic group. Panel tracks continuously employed workers across 12-month intervals with individual fixed effects. Source: CPS 2016-2022, men aged 25-54.

Figure 3. Employment Change by Education Groups

Panel A: Pandemic Period (2020) Only



Panel B: Full Period (2016-2024)



Notes: Figure shows employment change relative to February 2020 (Panel A) and January 2016 (Panel B) by education group. Panel B applies 3-month centered moving averages. Sample: Men aged 25-54, CPS 2016-2024.

3.2 Correlations with Employment Changes

To examine the composition mechanism directly, I compute correlations between monthly wage changes and changes in the employment-to-population ratio, following Solon, Barsky, and Parker (1994). The analysis relies on the structure of wage decompositions: changes in aggregate wages can be decomposed into a wage growth effect and a composition effect, as formalized by Daly, Hobijn, and Wiles (2012). Because this decomposition is additive, the correlation of aggregate wage changes with employment equals the sum of the correlations for each component.

Cross-sectional mean wages show a correlation of −0.26 with employment changes, indicating that measured wages tend to rise when employment falls. Demographic adjustment reduces this correlation to −0.22. The panel measure shows a correlation of essentially zero (−0.02).

The contrast between panel and cross-sectional correlations is informative. The panel measure isolates wage growth for continuously employed workers, corresponding to the wage growth effect in the decomposition framework. The near-zero panel correlation indicates that individual wage growth was largely unrelated to aggregate employment fluctuations during this period. Since the cross-sectional correlation equals the sum of the panel correlation and the composition correlation, and the panel correlation is approximately zero, the negative cross-sectional correlation arises primarily from the composition component. This confirms that the tendency for measured wages to rise when employment falls reflects compositional shifts rather than individual wage dynamics.

3.3 Within-Cell Wage Compression

The imputation bounds reveal an additional pattern in the data. Imputing wages for non-employed individuals at the 33rd percentile of their demographic cell yields higher measured growth at 2.8 percent than imputing at the 50th percentile, which is 2.6 percent. This ordering may seem counterintuitive, since the 33rd percentile assumption assigns lower wage levels to non-employed individuals. However, what matters for measured growth is not the wage level assigned in any single period but how each percentile of the within-cell distribution evolved over time.

The fact that assigning non-workers to lower within-cell positions yields higher measured growth indicates that the 33rd percentile of within-cell wage distributions grew faster than the 50th percentile over this period. This reveals wage compression within demographic cells: lower-wage workers within each age-education group experienced faster wage growth than higher-wage workers in the same cell.

This within-cell compression complements Autor, Dube, and McGrew's (2023) documentation of aggregate wage compression during the post-pandemic period. They show that wages at the bottom of the overall distribution grew faster than wages at the top, reversing approximately one-third of the four-decade increase in 90/10 log wage inequality. They attribute this compression to tight labor markets increasing competition for low-wage workers, with wage gains concentrated among workers who changed employers. My finding extends their analysis by showing that compression occurred not only across the aggregate wage distribution but also within more narrowly defined demographic cells. Even among workers with identical observable characteristics, those at lower points in the within-cell wage distribution experienced faster growth.

The wage compression pattern is consistent with the labor market competition mechanism proposed by Autor, Dube, and McGrew (2023). If tight labor markets enabled low-wage workers to move to better-paying positions, this would generate faster wage growth at lower percentiles regardless of demographic group. However, distinguishing whether this within-cell compression reflects job-changers bidding up wages at better-paying firms or incumbents receiving raises at their current employers would require data linking workers to their employers over time, which the CPS cannot provide.

3.4 Summary

Table 2 summarizes the main results. Panel A reports wage index values at key dates; Panel B reports cumulative growth and gaps from the panel estimate. Three patterns emerge. First, composition bias is substantial, and its direction depends on the comparison group. Cross-sectional measures understate wage gains for continuously employed workers by 9.9 percentage points while overstating gains relative to the full working-age population by approximately 2.7 percentage points. Second, observable demographics explain only about one-third of the composition effect, with the remainder reflecting selection on unobservable characteristics. Third, the within-cell wage distribution compressed over this period, with lower percentiles growing faster than higher percentiles.

Table 2: Cumulative Real Wage Growth of U.S. Men Aged 25-54 by Method, 2016 - 2022

Panel A: Real Wage Index Values

Method	Wage Index				
(Included Groups)	Feb 2016	Feb 2020	April 2020	June 2020	Feb 2022
Panel (Fixed Effects)	1.004	1.177	1.176	1.170	1.153
Cross-Sectional Mean	1.018	1.090	1.139	1.134	1.054
Cross-Sectional Median	1.005	1.094	1.158	1.158	1.056
Demographic-Adjusted	1.008	1.062	1.093	1.093	1.020
Imputed 50th Percentile	1.008	1.063	1.087	1.087	1.026
Imputed 33rd Percentile	1.007	1.067	1.082	1.082	1.028

Notes: Panel includes continuously employed workers matched across 12-month intervals. Cross-sectional measures include all employed workers each month. Demographic-adjusted holds age-education distribution constant at 2016 levels. Imputed measures include non-employed individuals assigned wages at the specified percentile of their demographic cell.

Panel B: Cumulative Growth and Gaps

Method	Cumulative Growth (%)	Gap from Panel (pp)
Panel (Fixed Effects)	15.3	—
Cross-Sectional Mean	5.4	-9.9
Cross-Sectional Median	5.6	-9.7
Demographic-Adjusted	2.0	-13.3
Imputed 50th Percentile	2.6	-12.7

Imputed 33rd Percentile	2.8	-12.5

Notes: All series normalized to January 2016 = 1.0 and smoothed using 3-month centered moving averages. Panel B reports cumulative growth from February 2016 to February 2022. "Gap from Panel" shows the difference between each method and the panel estimate; negative values indicate that the method understates wage growth relative to continuously employed workers. Sample: Men aged 25-54, CPS 2016-2022.

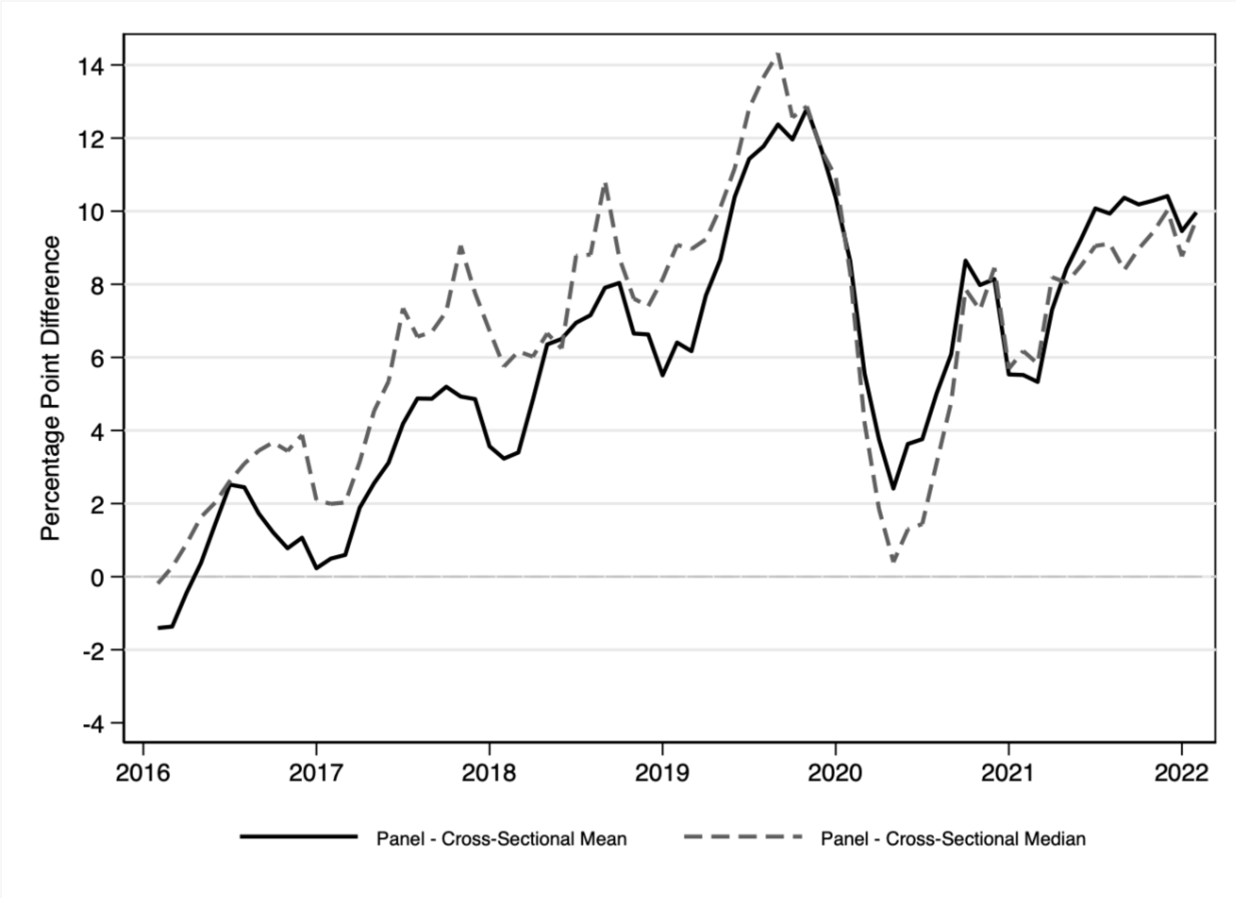
Section 4: Robustness

The main results compare panel and cross-sectional mean wages. Means can be sensitive to outliers in the wage distribution, and if extreme values drive the cross-sectional mean differently than the panel mean, the estimated composition effects could be overstated. To assess this concern, I examine whether the results hold when using medians instead of means.

Figure 4 shows that the gap between panel and cross-sectional medians (9.7 percentage points) is nearly identical to the gap using means (9.9 percentage points). Both measures exhibit similar temporal patterns: gaps widen during the pre-pandemic period, narrow during April–June 2020 when low-wage worker exits caused cross-sectional measures to spike, and widen again through 2022. The consistency between mean-based and median-based results indicates that outliers do not drive the main findings.

Several additional robustness concerns warrant discussion. First, differential attrition from the CPS panel could bias the panel estimates if workers who exit the sample between interviews differ systematically from those who remain. The CPS is designed to be representative at each interview, but selective attrition between outgoing rotations could affect the composition of matched pairs. Second, the demographic adjustment results depend on the cell definitions used. Finer cells by adding occupation or industry might capture more of the composition effect, while coarser cells might obscure it. Third, the restriction to men aged 25–54 ensures stable labor force attachment but limits generalizability. Women and younger workers exhibited different employment patterns during the pandemic (Doepke et al. 2022), and composition bias may operate differently for these groups. These extensions are beyond the scope of the current paper but represent important directions for future work.

Figure 4. Panel and cross-sectional wage gaps: mean vs. median real wages, 2016-2022



Note: Figure presents the gap between panel and cross-sectional wage indices using both means and medians. Positive values indicate that cross-sectional measures understate wage growth relative to continuously employed workers. All series smoothed using 3-month centered moving averages. Sample: Men aged 25–54, CPS 2016–2022.

Section 5: Conclusion

This paper makes three contributions to the literature on wage measurement and composition bias. First, I show that composition bias operates in opposing directions depending on the comparison group: cross-sectional measures understate wage gains for continuously employed workers while overstating gains relative to the full working-age population. This extends prior work documenting the existence of composition bias (Solon, Barsky, and Parker 1994; Cajner et al. 2020) by characterizing its multidirectional nature. Second, I quantify the share of composition effects attributable to observable versus unobservable characteristics, finding that demographics explain only about one-third of the gap between cross-sectional and panel estimates. Third, I document within-cell wage compression during the pandemic period, showing that even among workers with identical observable characteristics, those at lower percentiles of the wage distribution experienced faster growth. This finding complements Autor, Dube, and McGrew's (2023) documentation of aggregate compression.

These findings have practical implications for interpreting aggregate wage statistics during periods of employment disruption. When policymakers observed rising average wages in spring 2020, this pattern could have been mistaken for evidence that remaining workers were benefiting from tight labor markets. In fact, the measured increase largely reflected the exit of low-wage workers from the employed sample. Similarly, falling measured wages during the recovery may have signaled labor market weakness when they reflected the return of lower-wage workers to employment. For researchers using CPS data to study wage dynamics, these findings underscore the importance of panel methods or compositional adjustments when employment is volatile.

The analysis has several limitations. The CPS panel structure, with earnings observations separated by 12 months, prevents direct measurement of month-to-month wage changes during the acute pandemic period. Panel comparisons extend only through February 2022, precluding decomposition of composition effects during the subsequent inflationary period. Restricting the sample to prime-aged men ensures stable labor force attachment but limits generalizability to women and younger workers, who exhibited different employment patterns during the pandemic (Doepke et al. 2022). Finally, while panel estimates avoid selection bias from employment exits, they represent a selected sample of workers who retained employment throughout the observation period.

Several directions for future research emerge from this analysis. Extending the compositional decomposition to women would reveal whether their distinct pattern of pandemic job loss generated different compositional biases. Extending panel matching through 2024 would enable analysis of composition effects during the inflationary period. Most importantly, testing whether within-cell wage compression is concentrated among job-changers versus job-stayers would shed light on whether increased labor market competition explains the compression documented here. Addressing this question would require matched employer-employee data that can link individual wage trajectories to job transitions and firm characteristics, allowing researchers to distinguish between workers moving up the job ladder and incumbents receiving raises at their current employers.

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